Distributed Data Systems : Milestone 2

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Feedback

LACK OF RESEARCH STATEMENT:

- Exploring how different family conditions such as demographic and environmental factors impact the performance and wellbeing of students

LACK OF DATA WAREHOUSING:

- Acting as a central Repositor
- An Analytic platform for stakeholders

Merging of Data

- 2 data sets needed merging prior to be implemented into SQL
- Used Python in Visual Studio code to complete this task

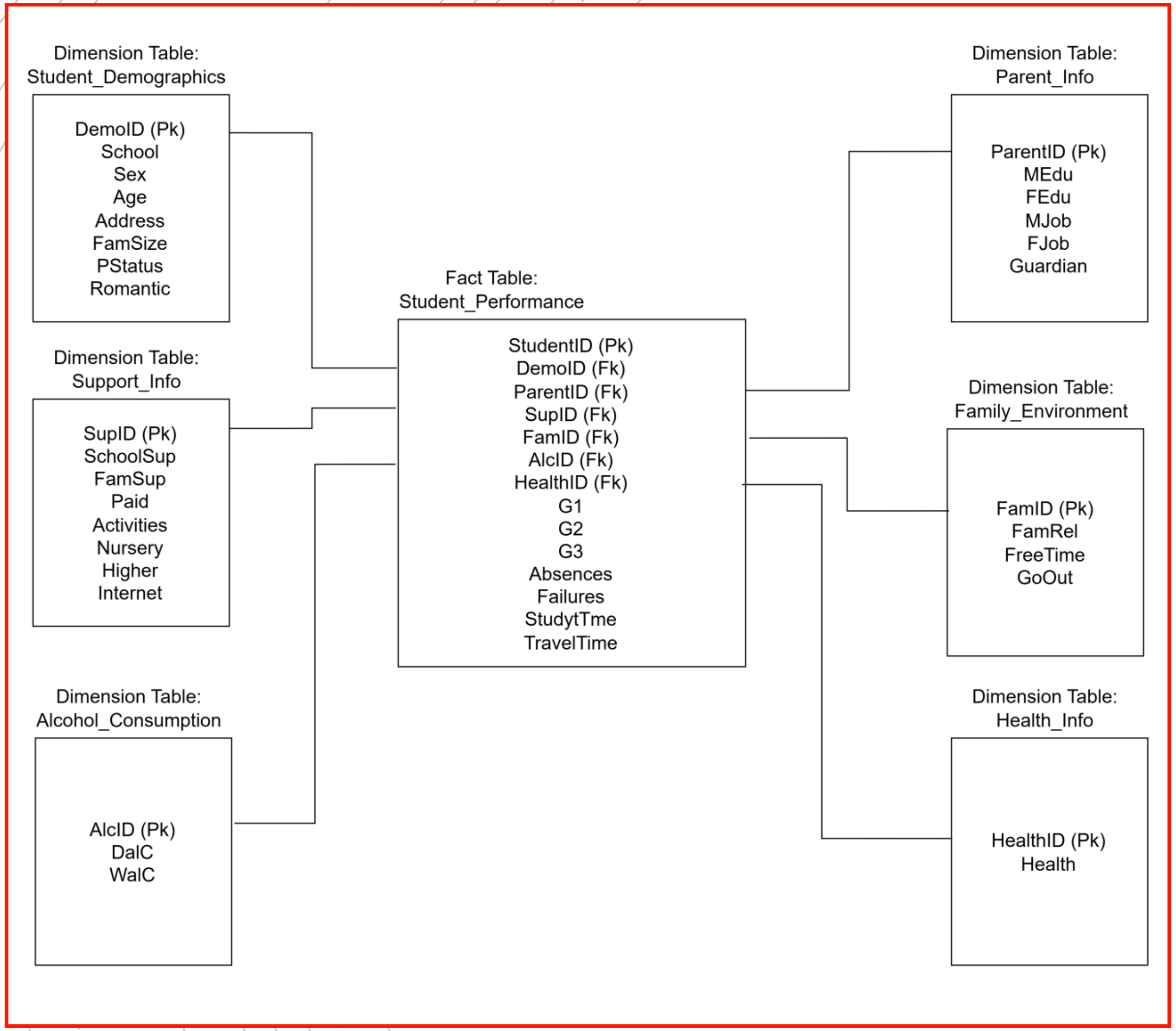
```
[28]: import pandas as pd

      d1 = pd.read_csv("student-mat.csv")
      d2 = pd.read_csv("student-por.csv")

      common_columns = [
          "school", "sex", "age", "address", "famsize", "Pstatus",
         "Medu", "Fedu", "Mjob", "Fjob", "reason", "nursery", "internet"
      ]

      d3 = pd.merge(d1, d2, on= common_columns)
      d3.to_csv("student_data.csv", index=False)
      len(d3)

[28]: 382
```



Implementing Tables into SQL

- Used Python in Visual Studio Code to Split data into unique tables
- Identified fact and dimensional tables based on schema in previous sections
- Created a connection to Root SQL within XXAMP

creating fact and dimension tables

```python
# Student Demographics
student_demographics = df[['school', 'sex', 'age', 'address', 'famsize', 'Pstatus', 'romantic_x']].drop_duplicates()
student_demographics['DemoID'] = range(1, len(student_demographics) + 1)

# Support Info
support_info = df[['schoolsup_x', 'famsup_x', 'paid_x', 'activities_x', 'nursery', 'higher_x', 'internet']].drop_duplicates()
support_info['SupID'] = range(1, len(support_info) + 1)

# Alcohol Consumption
alcohol_consumption = df[['Dalc_x', 'Walc_x']].drop_duplicates()
alcohol_consumption['AlcID'] = range(1, len(alcohol_consumption) + 1)

# Parent Info
parent_info = df[['Medu', 'Fedu', 'Mjob', 'Fjob', 'guardian_x']].drop_duplicates()
parent_info['ParentID'] = range(1, len(parent_info) + 1)

# Family Environment
family_environment = df[['famrel_x', 'freetime_x', 'goout_x']].drop_duplicates()
family_environment['FamID'] = range(1, len(family_environment) + 1)

# Health Info
health_info = df[['health_x']].drop_duplicates()
health_info['HealthID'] = range(1, len(health_info) + 1)
```

Merging Foreign keys and implementing into SQL

```python
# Merge foreign keys into the fact table
fact_table = df.merge(student_demographics, on=['school', 'sex', 'age', 'address', 'famsize', 'Pstatus', 'romantic_x']) \
                .merge(support_info, on=['schoolsup_x', 'famsup_x', 'paid_x', 'activities_x', 'nursery', 'higher_x', 'internet']) \
                .merge(alcohol_consumption, on=['Dalc_x', 'Walc_x']) \
                .merge(parent_info, on=['Medu', 'Fedu', 'Mjob', 'Fjob', 'guardian_x']) \
                .merge(family_environment, on=['famrel_x', 'freetime_x', 'goout_x']) \
                .merge(health_info, on=['health_x'])

# Select columns for fact table
fact_table = fact_table[['DemoID', 'ParentID', 'SupID', 'FamID', 'AlcID', 'HealthID', 'G1_x', 'G2_x', 'G3_x', 'absences_x', 'failures_x', 'studytime_x', 'traveltime_x']]
```

```python
# Create a connection to MySQL
engine = create_engine('mysql+pymysql://root@localhost/distributed data systems project')


# Test connection
engine.connect()
```

```
<sqlalchemy.engine.base.Connection at 0x22844babfd0>
```

```python
student_demographics.to_sql('Student_Demographics', engine, if_exists='replace', index=False)
support_info.to_sql('Support_Info', engine, if_exists='replace', index=False)
alcohol_consumption.to_sql('Alcohol_Consumption', engine, if_exists='replace', index=False)
parent_info.to_sql('Parent_Info', engine, if_exists='replace', index=False)
family_environment.to_sql('Family_Environment', engine, if_exists='replace', index=False)
health_info.to_sql('Health_Info', engine, if_exists='replace', index=False)
fact_table.to_sql('student_performance', engine, if_exists='replace', index=False)
```

Python

Testing

Conducted within SQL to ensure Data Loaded Properly

```
SELECT * FROM student_performance;
```

Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

1 > >> Show all | Number of rows: 25 Filter rows: Search this table

Extra options

DemoID	ParentID	SupID	FamID	AlcID	HealthID	G1_x	G2_x	G3_x	absences_x	failures_x	studytime_x	traveltime_x
1	1	1	1	1	1	5	6	6	6	0	2	2
49	66	9	12	7	1	10	8	9	10	0	1	1
62	139	4	12	1	1	9	10	0	0	0	3	1
87	108	38	12	2	1	8	7	8	7	1	1	1
2	147	8	12	1	1	13	12	12	6	0	3	1
2	56	8	12	1	1	15	12	14	0	0	3	2
75	150	8	12	1	1	14	13	14	0	0	4	1
56	68	13	12	8	1	9	12	11	3	0	3	1
37	90	4	12	3	1	8	8	8	10	0	4	1
75	159	10	12	3	1	11	10	10	4	0	3	1
96	167	51	12	2	1	14	12	12	3	0	1	3
2	2	2	2	1	1	5	5	6	4	0	2	1
18	30	6	2	1	1	13	14	14	0	0	1	2
2	116	13	2	1	1	8	8	9	0	0	2	1
48	127	15	2	4	1	12	11	12	2	0	2	1
33	61	30	4	4	1	10	12	12	2	0	1	1
90	49	41	4	3	1	12	11	12	0	0	2	2
9	56	6	10	1	1	10	10	10	0	0	4	1
20	7	21	10	3	1	7	10	11	25	1	2	2
16	113	4	10	3	1	11	12	11	2	0	1	1
51	114	2	10	4	1	8	7	9	6	0	2	1
2	123	5	10	9	1	7	7	8	3	0	1	2
86	61	13	34	1	1	11	11	10	2	0	2	2
69	57	44	34	1	1	11	11	11	2	0	2	2
74	3	50	11	1	1	12	12	12	1	0	4	2

```sql
SELECT
    student_demographics.School,
    student_demographics.Sex,
    support_info.schoolsup_x AS School_Support,
    support_info.famsup_x AS Family_Support,
    AVG(student_performance.G3_x) AS Average_Final_Grade
FROM
    Student_Performance
JOIN
    Student_Demographics ON student_performance.DemoID = student_demographics.DemoID
JOIN
    Support_Info  ON student_performance.SupID = support_info.SupID
GROUP BY
    student_demographics.School, student_demographics.Sex, support_info.SchoolSup_x, support_info.FamSup_x
```

School	Sex	School_Support	Family_Support	Average_Final_Grade 1
GP	M	no	no	11.3714
MS	M	no	yes	11.1667
GP	M	no	yes	11.1463
GP	F	yes	no	10.5455
GP	F	no	no	10.4444
GP	M	yes	no	10.3333
GP	F	no	yes	9.8932
GP	M	yes	yes	9.2500
MS	F	no	yes	9.2000
MS	M	no	no	9.1818
MS	F	no	no	9.1538
GP	F	yes	yes	9.0400

SQL Queries

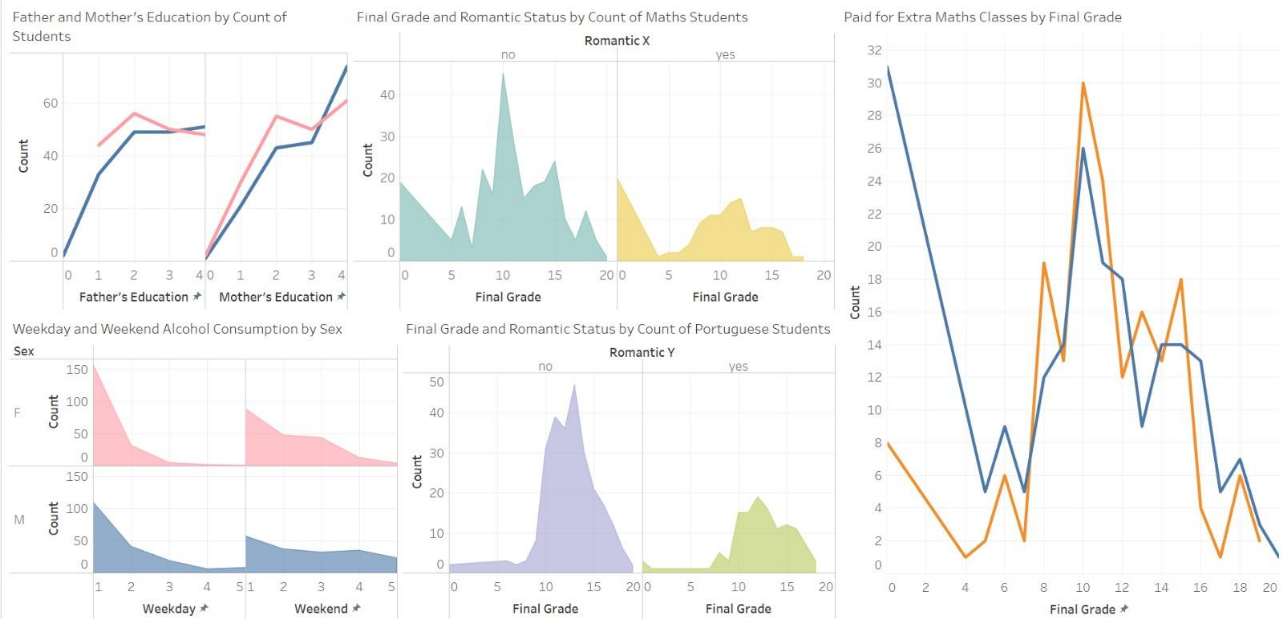


Tableau Visualisations

Any Question